

ALFIDO TECH

Cloud Computing Internship

TASK 4 COMPLETION REPORT

Deploy a Docker Container on Cloud VM

Submitted By	Saniya Shaikh
AWS Account	785446018888
Region	US East-1 (N. Virginia)
Instance ID	i-08b8a2d537ed464be
Instance IP	172.31.34.57 (Private) 184.72.91.111 (Public)
Docker Version	29.4.1 (build 055a478)
Container Image	my-nginx-app:latest (based on nginx:latest)
Container Name	mycontainer
Port Mapping	0.0.0.0:80 → Container Port 80 / TCP
Public Link	http://184.72.91.111
Date	April 26, 2026
Status	☒ COMPLETED

Step 1. Install Docker on EC2 VM

Docker was installed on the EC2 Ubuntu instance (ip-172-31-34-57) using the official Docker installation procedure. The service was started, enabled on boot, and verified with systemctl. Docker version 29.4.1 (build 055a478) was confirmed active and running since Apr 26 2026 15:59:04 UTC.

Commands executed:

```
sudo apt update -y
sudo apt install -y ca-certificates curl gnupg

# Add Docker GPG key and repository

sudo apt install -y docker-ce docker-ce-cli containerd.io

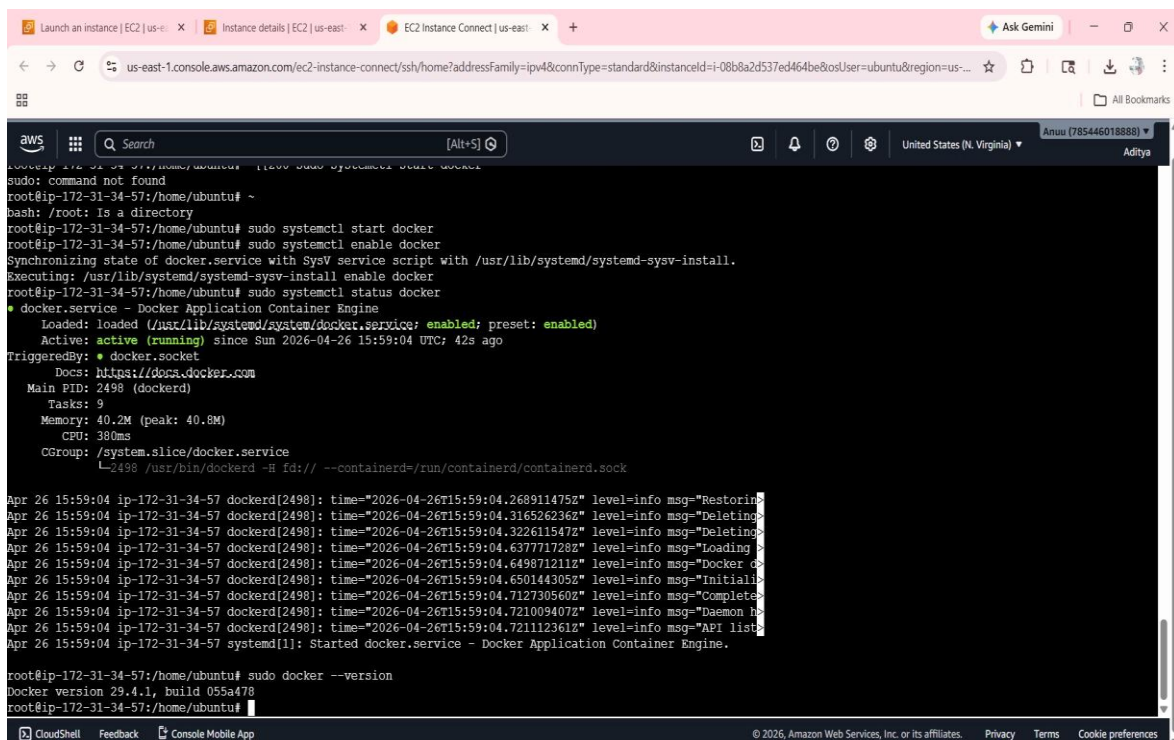
sudo systemctl start docker

sudo systemctl enable docker

sudo systemctl status docker

sudo docker --version

# Output: Docker version 29.4.1, build 055a478
```



```
Launch an instance | EC2 | us-east-1 | Instance details | EC2 | us-east-1 | EC2 Instance Connect | us-east-1 | + | Ask Gemini | - | x
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&iinstanceId=i-08b8a2d537ed464be8&osUser=ubuntu&region=us-east-1
AWS CloudShell
Search [Alt+S]
United States (N. Virginia) Annu (785446018888) Aditya
root@ip-172-31-34-57:/home/ubuntu# sudo apt update -y
root@ip-172-31-34-57:/home/ubuntu# sudo apt install -y ca-certificates curl gnupg
# Add Docker GPG key and repository
root@ip-172-31-34-57:/home/ubuntu# sudo apt install -y docker-ce docker-ce-cli containerd.io
root@ip-172-31-34-57:/home/ubuntu# sudo systemctl start docker
root@ip-172-31-34-57:/home/ubuntu# sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
root@ip-172-31-34-57:/home/ubuntu# sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-26 15:59:04 UTC; 42s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2498 (dockerd)
      Tasks: 9
    Memory: 40.2M (peak: 40.8M)
       CPU: 380ms
    CGroup: /system.slice/docker.service
            └─2498 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.268911475Z" level=info msg="Restoring
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.316526236Z" level=info msg="Deleting
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.322611547Z" level=info msg="Deleting
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.637771728Z" level=info msg="Loading
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.649871211Z" level=info msg="Docker co
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.650144305Z" level=info msg="Initiali
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.712730560Z" level=info msg="Comple
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.721009407Z" level=info msg="Daemon
Apr 26 15:59:04 ip-172-31-34-57 dockerd[2498]: time="2026-04-26T15:59:04.721112361Z" level=info msg="API list
Apr 26 15:59:04 ip-172-31-34-57 systemd[1]: Started docker.service - Docker Application Container Engine.

root@ip-172-31-34-57:/home/ubuntu# sudo docker --version
Docker version 29.4.1, build 055a478
root@ip-172-31-34-57:/home/ubuntu#
```

Figure 1 – Docker v29.4.1 installed and active (running) on EC2 instance ip-172-31-34-57 | Confirmed via: `sudo docker --version`

Step 2. Create Dockerfile & Build Docker Image

A project directory ~/myapp was created containing a custom index.html and a Dockerfile. The Dockerfile uses nginx:latest as the base image and copies the HTML file into the Nginx web root. The image was built successfully in 4.1 seconds across 7 build steps, pulling all required layers from docker.io.

Dockerfile contents:

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html

EXPOSE 80
```

Build command:

```
mkdir ~/myapp && cd ~/myapp
# (Create index.html and Dockerfile in this directory)

sudo docker build -t my-nginx-app .

# Output: [+] Building 4.1s (7/7) FINISHED   docker:default
```

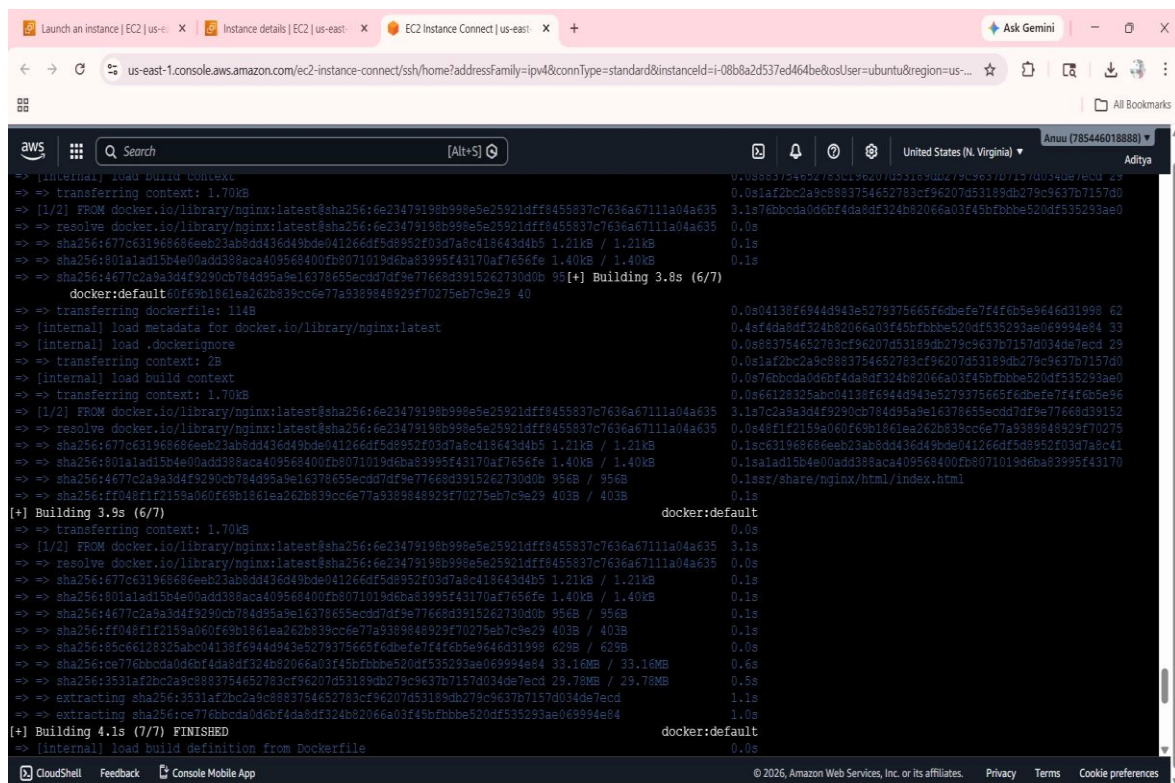


Figure 2 – Docker image build completed: [+] Building 4.1s (7/7) FINISHED | nginx:latest pulled and custom HTML layer copied successfully

Step 3. Verify Docker Image Created

After the build, the docker images command confirmed that my-nginx-app:latest was successfully created with an image ID of 3e84161348c6, occupying 237 MB disk space with 63 MB content size. The hello-world:latest test image (25.9 KB) was also present from the initial Docker installation verification.

Verification command:

```
sudo docker images

# OUTPUT:

# IMAGE                                ID                                DISK USAGE    CONTENT SIZE
# hello-world:latest                  f9078146db2e                      25.9kB        9.49kB
# my-nginx-app:latest                 3e84161348c6                      237MB         63MB
```

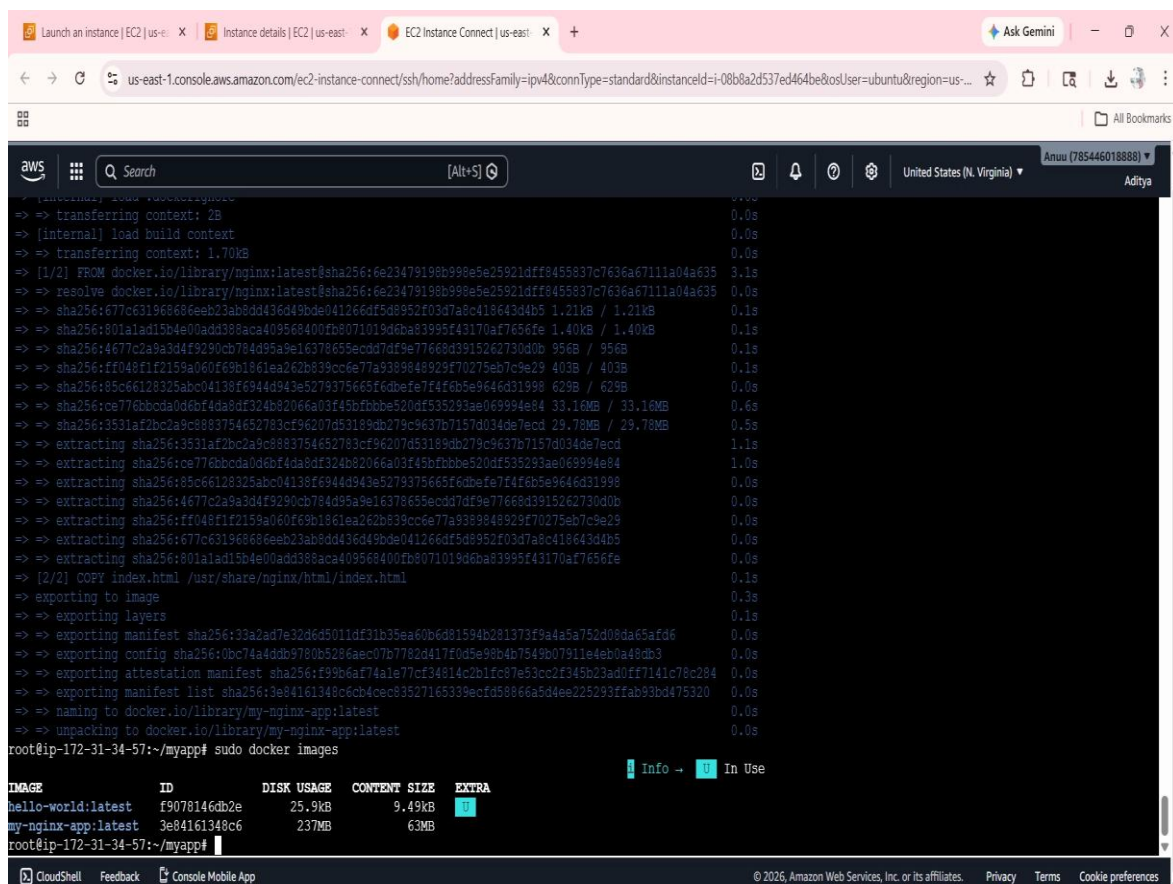


Figure 3 – Docker images list: my-nginx-app:latest (237MB) successfully created | Image ID: 3e84161348c6

Step 4. Run Docker Container & Expose Port 80

The container named mycontainer was launched in detached mode (-d) with port 80 on the host mapped to port 80 in the container (-p 80:80). The docker ps command confirmed the container is running with STATUS 'Up 18 seconds' and ports 0.0.0.0:80→80/tcp and [::]:80→80/tcp exposed. A local curl test was also performed to verify the container serves content correctly.

Commands executed:

```
# Run container in detached mode with port mapping
sudo docker run -d --name mycontainer -p 80:80 my-nginx-app

# Output:
4f968a373f23d0562d1ad86cb0498ef75dd74e09e32e0b30c32e15d0794b2b73

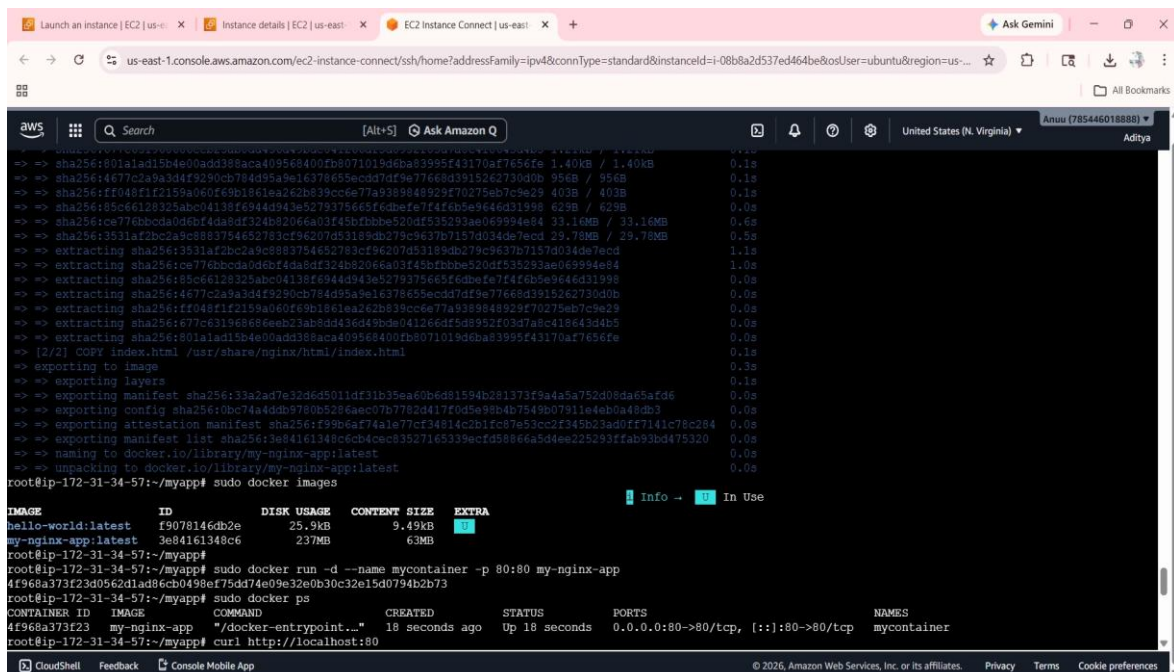
# Verify container is running

sudo docker ps

# CONTAINER ID   IMAGE                                STATUS      PORTS
# 4f968a373f23   my-nginx-app                        Up 18 seconds    0.0.0.0:80->80/tcp

# Local test

curl http://localhost:80
```



The screenshot shows an AWS CloudShell terminal window with the following commands and output:

```
root@ip-172-31-34-57:/myapp# sudo docker images
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
hello-world:latest                  f9078146db2e                      25.9kB      9.49kB
my-nginx-app:latest                 3e84161348c6                      237MB      63MB

root@ip-172-31-34-57:/myapp# sudo docker run -d --name mycontainer -p 80:80 my-nginx-app
4f968a373f23d0562d1ad86cb0498ef75dd74e09e32e0b30c32e15d0794b2b73

root@ip-172-31-34-57:/myapp# sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS      PORTS                                NAMES
4f968a373f23   my-nginx-app                        "/docker-entrypoint..." 18 seconds ago Up 18 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp  mycontainer

root@ip-172-31-34-57:/myapp# curl http://localhost:80
```

Figure 4 – Container 'mycontainer' running: Up 18 seconds | Ports: 0.0.0.0:80→80/tcp confirmed via sudo docker ps

Step 5. Live Container Output via Public IP in Browser

The Docker container's web application is publicly accessible through the EC2 instance's public IP address. The browser at <http://184.72.91.111> displays the custom 'Hello from Docker!' page, confirming the Nginx container is serving content correctly over port 80 to the internet. The page shows the container image (nginx:latest), the host platform (AWS EC2 Ubuntu VM), and the port mapping (Port 80 → Container Port 80 → Internet).

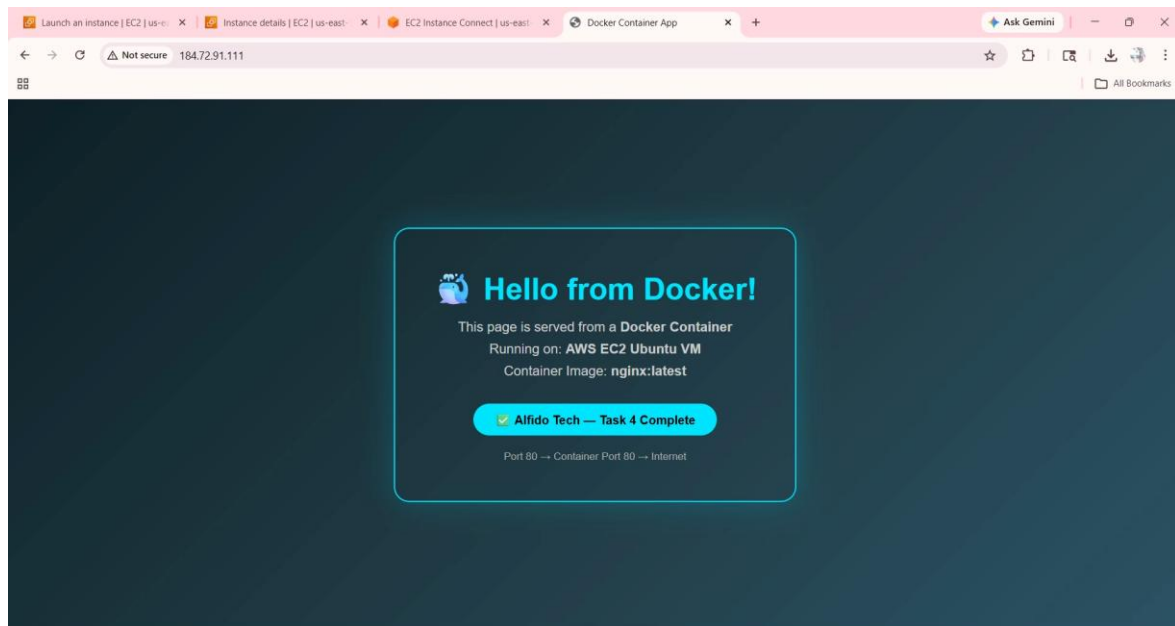


Figure 5 – Live 'Hello from Docker!' page accessible at <http://184.72.91.111> | Served by Nginx container on AWS EC2 Ubuntu VM — Task 4 Complete

Task Completion Summary

Requirement	Action Taken	Status
Install Docker on VM	Docker v29.4.1 installed, started & verified active/running	✓ Done
Run container using image (nginx)	my-nginx-app built from nginx:latest; mycontainer launched	✓ Done
Expose container port to internet	Port 80 mapped (0.0.0.0:80→80/tcp); Security Group open	✓ Done
Public link showing container output	http://184.72.91.111 live — Hello from Docker! page visible	✓ Done

Complete Docker Commands Reference

```
# — Step 1: Install Docker —————
sudo apt update -y

sudo apt install -y ca-certificates curl gnupg

sudo apt install -y docker-ce docker-ce-cli containerd.io

sudo systemctl start docker

sudo systemctl enable docker

sudo systemctl status docker

sudo docker -version


# — Step 2: Create project files —————
mkdir ~/myapp && cd ~/myapp

nano index.html          # Paste custom HTML content

nano Dockerfile          # FROM nginx:latest | COPY | EXPOSE 80


# — Step 3: Build Docker image —————
sudo docker build -t my-nginx-app .

sudo docker images
```

```
# — Step 4: Run container on port 80 —————  
sudo docker run -d --name mycontainer -p 80:80 my-nginx-app  
sudo docker ps  
curl http://localhost:80  
  
# — Step 5: Access in browser —————  
# Open: http://184.72.91.111  
  
# — Useful additional commands —————  
  
sudo docker logs mycontainer          # View container logs  
sudo docker stop mycontainer          # Stop the container  
sudo docker rm mycontainer            # Remove the container  
sudo docker rmi my-nginx-app          # Remove the image
```

Deliverables

1. Public Link / IP Showing Container Output:

<http://184.72.91.111>

2. Screenshot + Docker Command List:

- Docker v29.4.1 installed and running — systemctl status + docker --version (Figure 1)
- Docker image build completed — [+] Building 4.1s (7/7) FINISHED (Figure 2)
- Docker images list — my-nginx-app:latest (237MB, ID: 3e84161348c6) (Figure 3)
- Container running — sudo docker ps | mycontainer Up 18 seconds (Figure 4)
- Live browser output — Hello from Docker! at <http://184.72.91.111> (Figure 5)
- Complete Docker command list documented in Commands Reference section above